

# ISyE 6669

## Dantzig-Wolfe Decomposition

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Spring 2020

In the last module, we introduced the general framework of column generation and constraint generation for solving large scale linear optimization problems. We applied the column generation algorithm to the cutting stock problem. In this lecture, we study another application of the column generation technique, the so-called *Dantzig-Wolfe decomposition*. It is designed to deal with large scale LPs with special structures. We will first derive the Dantzig-Wolfe decomposition principle, then, see an example.

### 1 The Dantzig-Wolfe Decomposition

Consider the following LP:

$$\begin{aligned} (LP) \quad & \min \quad \mathbf{c}^T \mathbf{x} \\ & \text{s.t.} \quad \mathbf{D}\mathbf{x} = \mathbf{b}_0 \\ & \quad \mathbf{F}\mathbf{x} = \mathbf{b} \\ & \quad \mathbf{x} \geq \mathbf{0} \end{aligned}$$

where  $\mathbf{D}$  is a  $m_1 \times n$  matrix, and  $\mathbf{F}$  is a  $m_2 \times n$  matrix. So there are  $m_1 + m_2$  equality constraints and  $n$  variables.

#### Step 1: Separate the Constraint Set into Two

The above problem is already written in an explicit way that there are two sets of equality constraints. Suppose the first set  $\mathbf{D}\mathbf{x} = \mathbf{b}_0$  is harder to deal with. We can define a polyhedron  $P$  using the constraints  $\mathbf{F}\mathbf{x} = \mathbf{b}$  and  $\mathbf{x} \geq \mathbf{0}$  as follows:

$$P = \{\mathbf{x} \mid \mathbf{F}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Then, the original problem can be written in the following way, which singles out the hard constraint  $\mathbf{D}\mathbf{x} = \mathbf{b}_0$ :

$$\begin{aligned} (LP_2) \quad & \min \quad \mathbf{c}^T \mathbf{x} \\ & \text{s.t.} \quad \mathbf{D}\mathbf{x} = \mathbf{b}_0 \\ & \quad \mathbf{x} \in P \end{aligned}$$

This separation usually depends on the problem structure. We will talk more about this later.

## Step 2: Use the Extreme Points Representation (Key Idea!)

The following idea of reformulating  $(LP_2)$  will probably blow your mind. Just sit back and watch.

As we already discussed in earlier lectures, any point in a polytope (i.e., a bounded polyhedron) can be represented as a convex combination of its extreme points. Let us denote all the extreme points of  $P$  as  $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^N$ . Here each  $\mathbf{x}^i$  is a vector in  $\mathbb{R}^n$ , and the number of extreme points  $N$  can be huge.

**Example:** Here is an example. Think about how many extreme points a two-dimensional cube has. A cube in  $\mathbb{R}^2$  can be described as the set  $\{\mathbf{x} : 0 \leq x_i \leq 1, \forall i = 1, 2\}$ . The answer is 4. How many extreme points does a three-dimensional cube have? The answer is 8. How many extreme points does a  $n$ -dimensional cube have? The answer is  $2^n$ . When  $n = 10$ , the number is 1024. It gets big exponentially quickly as  $n$  grows.

However, we already know that, regardless of how large  $N$  is, we can always write any point  $\mathbf{x}$  in the polytope  $P$  as a convex combination of its extreme points as:

$$\begin{aligned}\mathbf{x} &= \sum_{i=1}^N \lambda_i \mathbf{x}^i \\ \sum_{i=1}^N \lambda_i &= 1 \\ \lambda_i &\geq 0, \forall i = 1, \dots, N.\end{aligned}\tag{1}$$

The weights  $\lambda_i$ 's depend on where the point  $\mathbf{x}$  is. Different  $\mathbf{x}$ 's will have different  $\lambda_i$ 's. But the extreme points  $\mathbf{x}^i$ 's are fixed for the polyhedron  $P$ .

Now comes the **key idea** of Dantzig and Wolfe: **In the problem  $(LP_2)$ , we can replace the variable  $\mathbf{x}$  by its convex representation (1). The new variables are  $\lambda_i$ 's.**

$(LP_2)$  now becomes

$$\begin{aligned}\min_{\lambda_1, \dots, \lambda_N} \quad & \mathbf{c}^T \left( \sum_{i=1}^N \lambda_i \mathbf{x}^i \right) \\ \text{s.t.} \quad & \mathbf{D} \left( \sum_{i=1}^N \lambda_i \mathbf{x}^i \right) = \mathbf{b}_0 \\ & \sum_{i=1}^N \lambda_i = 1 \\ & \lambda_i \geq 0, \forall i = 1, \dots, N.\end{aligned}$$

Multiply  $\mathbf{c}$  and  $\mathbf{D}$  into the parenthesis, we have

$$\begin{aligned}
 (MP) \quad & \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\
 & \text{s.t.} \quad \sum_{i=1}^N \lambda_i (\mathbf{D} \mathbf{x}^i) = \mathbf{b}_0 \\
 & \quad \sum_{i=1}^N \lambda_i = 1 \\
 & \quad \lambda_i \geq 0, \forall i = 1, \dots, N.
 \end{aligned}$$

(MP) stands for the master problem. (MP) is exactly equivalent to the original problem (LP), namely, if  $\lambda_1, \dots, \lambda_N$  is optimal for (MP), then we can use (1) to recover an optimal solution  $\mathbf{x}$  for (LP).

### Step 3: Apply Column Generation to (MP)

Let us look at the master problem (MP). It has  $N$  variables  $\lambda_1, \lambda_2, \dots, \lambda_N$ , which can be a very large number, and  $(m_1 + 1)$  equality constraints. Note that  $\mathbf{c}^T \mathbf{x}^i$  is a scalar number, and  $\mathbf{D} \mathbf{x}^i$  is a vector in  $\mathbb{R}^{m_1}$ . We are in a perfect situation to apply the Column Generation Algorithm to solve (MP).

1. Choose a subset  $I$  of columns and variables. Solve the restricted master problem:

$$\begin{aligned}
 (RMP) \quad & \min_{\lambda_i, \forall i \in I} \sum_{i \in I} \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\
 & \text{s.t.} \quad \sum_{i \in I} \lambda_i (\mathbf{D} \mathbf{x}^i) = \mathbf{b}_0 \quad \leftarrow \hat{\mathbf{y}} \\
 & \quad \sum_{i \in I} \lambda_i = 1 \quad \leftarrow \hat{r} \\
 & \quad \lambda_i \geq 0, \forall i \in I.
 \end{aligned}$$

Obtain an optimal solution  $\hat{\boldsymbol{\lambda}}$  of (RMP), and an optimal dual solution  $\hat{\mathbf{y}}$  and  $\hat{r}$ .

2. Check the optimality of  $\hat{\boldsymbol{\lambda}}$  using the dual information (recall Notes 6). Here is a good exercise to write down the reduced cost. See if you can do it without looking at the solution below.

Find the minimum reduced cost:

$$Z = \min_{i=1, \dots, N} \left\{ (\mathbf{c}^T \mathbf{x}^i) - \hat{\mathbf{y}}^T (\mathbf{D} \mathbf{x}^i) - \hat{r} \right\} \quad (2)$$

If  $Z \geq 0$ , then  $\hat{\boldsymbol{\lambda}}$  is optimal for (MP). Terminate the algorithm. Otherwise, add the column with the minimum reduced cost to (RMP), and continue.

The above Column Generation Algorithm is clear. However, to make it effective, we need to be able to find the minimum reduced cost efficiently. Enumerating all the extreme points in (2) will

be too slow, plus we do not even know all the extreme points. We need to be able to *generate* them on the fly.

Now comes the second **key idea** of Dantzig and Wolfe, which is essentially the same as what we did in the pricing subproblem for the cutting stock problem. Namely, **enumerating through all the extreme points of  $P$  in (2) is equivalent to minimizing the reduced cost over the polytope  $P$  directly!**

$$\begin{aligned}
Z &= \min_{i=1,\dots,N} \left\{ (\mathbf{c}^T \mathbf{x}^i) - \hat{\mathbf{y}}^T (\mathbf{D} \mathbf{x}^i) - \hat{r} \right\} \\
&= \min_{\mathbf{x}} \left\{ (\mathbf{c}^T \mathbf{x}) - \hat{\mathbf{y}}^T (\mathbf{D} \mathbf{x}) - \hat{r} \right\} \\
&\quad \text{s.t. } \mathbf{x} \in P \\
&= \min_{\mathbf{x}} \left\{ (\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D}) \mathbf{x} \right\} - \hat{r} \\
&\quad \text{s.t. } \mathbf{F} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}
\end{aligned}$$

Note that the dual variable  $\hat{r}$  is fixed, so we can pull it outside of the minimization problem. This step here is a reverse application of the extreme point representation: going from the extreme point representation of a point back to its polytope representation.

Now an efficient Column Generation Algorithm can be written as:

1. Choose a subset  $I$  of columns and variables. Solve the restricted master problem:

$$\begin{aligned}
(RMP) \quad & \min_{\lambda_i, \forall i \in I} \sum_{i \in I} \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\
& \text{s.t. } \sum_{i \in I} \lambda_i (\mathbf{D} \mathbf{x}^i) = \mathbf{b}_0 \quad \leftarrow \hat{\mathbf{y}} \\
& \sum_{i \in I} \lambda_i = 1 \quad \leftarrow \hat{r} \\
& \lambda_i \geq 0, \forall i \in I.
\end{aligned}$$

Obtain an optimal solution  $\hat{\lambda}$  of (RMP). And the optimal dual solution  $\hat{\mathbf{y}}$  and  $\hat{r}$ .

2. To check the optimality of  $\hat{\lambda}$ , we solve the following subproblem (SP):

$$\begin{aligned}
(SP) \quad & w = \min \quad (\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D}) \mathbf{x} \\
& \text{s.t. } \mathbf{F} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}.
\end{aligned}$$

If  $w \geq \hat{r}$ , then  $\hat{\lambda}$  is optimal for (MP). Terminate the column generation algorithm.

Otherwise, the optimal basic feasible solution of the above problem is an extreme point of the polyhedron  $P$ . Denote it as  $\mathbf{x}^i$ . Add the column  $\begin{bmatrix} \mathbf{D} \mathbf{x}^i \\ 1 \end{bmatrix}$  to the constraint matrix in restricted master problem, and add  $\mathbf{c}^T \mathbf{x}^i$  to the cost coefficient.

**Remarks:**

- Note that the master problem (MP) can be written in the matrix form as follows:

$$\begin{aligned}
(MP) \quad & \min_{\lambda_1, \dots, \lambda_N} \begin{bmatrix} \mathbf{c}^T \mathbf{x}^1 \\ \mathbf{c}^T \mathbf{x}^2 \\ \vdots \\ \mathbf{c}^T \mathbf{x}^N \end{bmatrix}^T \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_N \end{bmatrix} \\
& \text{s.t.} \quad \begin{bmatrix} \mathbf{D}\mathbf{x}^1 & \mathbf{D}\mathbf{x}^2 & \dots & \mathbf{D}\mathbf{x}^N \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_N \end{bmatrix} = \begin{bmatrix} \mathbf{b}_0 \\ 1 \end{bmatrix} \\
& \lambda_i \geq 0, \forall i = 1, \dots, N.
\end{aligned}$$

The restricted master problem (RMP) involves a subset of variables and columns of (MP).

- The dual variables  $\hat{\mathbf{y}}$  and  $\hat{r}$  can be obtained from computing  $\mathbf{c}_B^T \mathbf{B}^{-1}$  as we did in the simplex method. In particular,  $\hat{\mathbf{y}}$  is the first  $m_1$  components of  $\mathbf{c}_B^T \mathbf{B}^{-1}$ , and  $\hat{r}$  is the last component of  $\mathbf{c}_B^T \mathbf{B}^{-1}$ . Here  $\mathbf{c}_B$  is the vector of coefficients of  $\lambda_i$ 's corresponding to the basic variables, and  $\mathbf{B}$  is the basis matrix of the big matrix  $\begin{bmatrix} \mathbf{D}\mathbf{x}^1 & \mathbf{D}\mathbf{x}^1 & \dots & \mathbf{D}\mathbf{x}^N \\ 1 & 1 & \dots & 1 \end{bmatrix}$ .

## 2 A Numerical Example

We now consider a numerical example and go through the details of the algorithm. This will give us some hands-on experience with the algorithm.

$$\begin{aligned}
\min \quad & -4x_1 - x_2 - 6x_3 \\
\text{s.t.} \quad & 3x_1 + 2x_2 + 4x_3 = 17 \\
& 1 \leq x_1 \leq 2 \\
& 1 \leq x_2 \leq 2 \\
& 1 \leq x_3 \leq 2
\end{aligned}$$

We divide the constraints into two groups: The first group consists of the constraint  $\mathbf{D}\mathbf{x} = \mathbf{b}_0$  where  $\mathbf{D} = [3 \ 2 \ 4]$  and  $\mathbf{b}_0 = 17$ . The second group is the constraint  $\mathbf{x} \in P$ , where the polyhedron  $P = \{\mathbf{x} \in \mathbb{R}^3 : 1 \leq x_i \leq 2, \forall i = 1, 2, 3\}$ .  $P$  is a 3-dimensional cube, which has eight extreme points  $\mathbf{x}^1, \dots, \mathbf{x}^8$ ; it is bounded and has no extreme rays.

The master problem (MP) is written as

$$\begin{aligned}
 (MP) \quad & \min_{\lambda_1, \dots, \lambda_8} \sum_{i=1}^8 \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\
 \text{s.t.} \quad & \sum_{i=1}^8 \lambda_i (\mathbf{D} \mathbf{x}^i) = 17 \\
 & \sum_{i=1}^8 \lambda_i = 1 \\
 & \lambda_i \geq 0, \forall i = 1, \dots, 8.
 \end{aligned}$$

Note that (MP) has two equality constraints. So let us pick two columns, or equivalently choose two extreme points of  $P$ , to start the column generation algorithm. Pick  $\mathbf{x}^1 = (2, 2, 2)$  and  $\mathbf{x}^2 = (1, 1, 2)$ , and the corresponding convex weights  $\lambda_1, \lambda_2$ . Note that here the specific indices do not matter. We can always order the extreme points so that the first extreme point is  $(2, 2, 2)$  and the second is  $(1, 1, 2)$ . In class we used another ordering where the first extreme point is  $(1, 1, 1)$  and the eighth is  $(2, 2, 2)$ . But this ordering is not material. We have

$$\begin{aligned}
 \mathbf{c}^T \mathbf{x}^1 &= [-4 \ -1 \ -6] \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = -22, \quad \mathbf{c}^T \mathbf{x}^2 = [-4 \ -1 \ -6] \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = -17 \\
 \mathbf{D} \mathbf{x}^1 &= [3 \ 2 \ 4] \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = 18, \quad \mathbf{D} \mathbf{x}^2 = [3 \ 2 \ 4] \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 13.
 \end{aligned}$$

So the restricted master problem (RMP) can be written explicitly as

$$\begin{aligned}
 (RMP) \quad & \min_{\lambda_1, \lambda_2} -22\lambda_1 - 17\lambda_2 \\
 \text{s.t.} \quad & 18\lambda_1 + 13\lambda_2 = 17 \\
 & \lambda_1 + \lambda_2 = 1 \\
 & \lambda_1, \lambda_2 \geq 0.
 \end{aligned}$$

The basis matrix and its inverse are

$$\mathbf{B} = \begin{bmatrix} 18 & 13 \\ 1 & 1 \end{bmatrix} \quad \mathbf{B}^{-1} = \begin{bmatrix} 0.2 & -2.6 \\ -0.2 & 3.6 \end{bmatrix}$$

The optimal solution is  $\begin{bmatrix} \hat{\lambda}_1 \\ \hat{\lambda}_2 \end{bmatrix} = \mathbf{B}^{-1} \begin{bmatrix} 17 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$ . Form the dual variable:

$$\begin{bmatrix} \hat{\mathbf{y}}^T & \hat{r} \end{bmatrix} = \mathbf{c}_B^T \mathbf{B}^{-1} = [-22 \ -17] \mathbf{B}^{-1} = [-1 \ -4].$$

To compute the minimum reduced cost, we form the following subproblem:

$$\begin{aligned}
 (SP) \quad & \min \left( \mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} \right) \mathbf{x} \\
 \text{s.t.} \quad & \mathbf{x} \in P
 \end{aligned}$$

which is written more explicitly (note  $\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} = [-4, -1, -6] - (-1)[3, 2, 4] = [-1, 1, -2]$ ):

$$\begin{aligned} w = \min \quad & -x_1 + x_2 - 2x_3 \\ \text{s.t.} \quad & 1 \leq x_1 \leq 2, 1 \leq x_2 \leq 2, 1 \leq x_3 \leq 2. \end{aligned}$$

The optimal solution is  $\mathbf{x}^3 = [2, 1, 2]$ , which is another extreme point of the cube  $P$ . The optimal cost  $w = -5$ , which is less than  $\hat{r} = -4$ , therefore, the reduced cost of  $\lambda_3$  is  $-1$ , this variable enters into the restricted problem. The associated column is  $[\mathbf{D}\mathbf{x}^3, 1] = [16, 1]$ , and the associated cost coefficient is  $\mathbf{c}^T \mathbf{x}^3 = -21$ . The updated (RMP) is

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \lambda_2, \lambda_3} \quad -22\lambda_1 - 17\lambda_2 - 21\lambda_3 \\ \text{s.t.} \quad & 18\lambda_1 + 13\lambda_2 + 16\lambda_3 = 17 \\ & \lambda_1 + \lambda_2 + \lambda_3 = 1 \\ & \lambda_1, \lambda_2, \lambda_3 \geq 0. \end{aligned}$$

The optimal solution is  $\hat{\lambda}_1 = 0.5, \hat{\lambda}_2 = 0, \hat{\lambda}_3 = 0.5$ . The optimal basis matrix and its inverse are

$$\mathbf{B} = \begin{bmatrix} 18 & 16 \\ 1 & 1 \end{bmatrix} \quad \mathbf{B}^{-1} = \begin{bmatrix} 0.5 & -8 \\ -0.5 & 9 \end{bmatrix}$$

The dual variables are

$$[\hat{\mathbf{y}}, \hat{r}] = \mathbf{c}_B^T \mathbf{B}^{-1} = [-22, -21] \mathbf{B}^{-1} = [-0.5, -13].$$

To check the optimality of the solution  $\hat{\lambda}_i$ 's, we need to form the subproblem.

$$\mathbf{c}^T - \hat{\mathbf{y}}^T \mathbf{D} = [-4, -1, -6] - (-0.5)[3, 2, 4] = [-2.5, 0, -4]$$

The subproblem is given as

$$\begin{aligned} w = \min \quad & -2.5x_1 - 4x_3 \\ \text{s.t.} \quad & 1 \leq x_1 \leq 2, 1 \leq x_2 \leq 2, 1 \leq x_3 \leq 2. \end{aligned}$$

An optimal solution is  $\mathbf{x}^4 = [2, 2, 2]$  with the optimal cost  $w = -13$ . Note that  $\hat{r} = -13$ . So  $w = \hat{r}$  which means the current solution is optimal for the master problem. In terms of the variables  $x_i$ , the optimal solution is

$$\mathbf{x}^* = 0.5\mathbf{x}^1 + 0.5\mathbf{x}^3 = 0.5 \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + 0.5 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1.5 \\ 2 \end{bmatrix}.$$